MIL-P-11635C(EA)
22 July 1987
SUPERSEDING
MIL-P-11635B(MU)
18 September 1968

MILITARY SPECIFICATION

PROJECTILE, 57 MM, SMOKE, WP, M308A1; FILLING ASSEMBLY

This specification is approved for use within the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the white phosphorus (WP) filling and assembly of the M308A1 projectile, without explosive components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Ammunition Lot Numbering.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

9215578 - Projectile, 57 MM, Smoke, WP, M308A1; Filling Assembly.
10543043 - Projectile, 57 MM, Smoke, WP, M308A1; Metal Parts Assembly.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 10543043, or on subsidiary drawings, shall conform to the drawings and to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Filling. The projectile, clean and free from foreign matter, shall be filled with molten WP conforming to MIL-C-215, at a temperature of 140° ± 10°F. When water is in contact with the WP during the filling operation, the water used shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The depth of water remaining on the surface of the WP in the filled and closed projectile shall not exceed 1/8 inch. After testing, the filling shall be solidified with the projectile in a filling-end-up position (see 6.2 and 6.3).

3.3 Assembly. Closing, coating, and marking of the projectile shall be in accordance with Drawing 9215578. The fuze and base threads shall be coated with silicone compound as specified on the drawing.

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3.4 Filling height. There shall be a 10.5 to 17 percent void, by total volume, existing above the surface of the water over the WP in the filled and closed projectile. The projectile shall be radiographed in a vertical, filling-end-up position while the termperature of the WP is 140° ± 10°F when tested as specified in 4.2.4.1 (see 6.3).

3.5 Cleaning after closing. The exterior surface of the filled and closed projectile and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, WP or other foreign matter (see 6.4).

3.6 Leakage. The closed and cleaned projectile shall not leak when maintained at a temperature of 160° ± 5°F, for a period of 8 hours. Alternate method: the projectile shall not leak when maintained at a temperature of 207° to 212°F for a minimum period of 15 minutes. The projectile shall be tested as specified in 4.2.4.2.

3.7 Workmanship. The projectile shall be free from foreign matter (oil, grease, and contamination) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of projectiles, of one interfix number, filled by the same filling installation, under essentially the same conditions, in one work week. However, no more than one manufacturer or interfix number of body assemblies or burster wells shall be represented in any one lot. Lotting shall be in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample projectiles shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each filled and closed projectile shall be inspected for the critical defects listed in the classification of defects.

4.2.3.3 Classification of defects.

(a) Projectile, 57 mm, smoke, WP, M308A1; metal parts assembly (Drawing 10543043).

Categories	Defects	Acceptance standards
Critical:	100 percent inspection	
1	WP leakage	4.2.4.2
2	WP contamination (3.5)	
Major:	AQL 1.0 percent defective	
101	Filling excessive	4.2.4.1
102	Excess water	4.2.4.1
103	Filling insufficient	4.2.4.1
104	Burster casing assembly improperly seated	
105	Bourrelet diameter incorrect after protective finish and marking	Ring gauge
106	Protective finish missing	
107	Marking incorrect	
108	Threads damaged	
109	Workmanship (3.7)	
110	Rotating band groove damaged	

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Minor: AQL 2.5 percent defective

201	Crimping grooves damaged
202	Marking missing or illegible

(b) Packaging inspection (see section 5).

Categories	Defects
Critical:	None defined
Major: AQL 4.0 percent defective	
101	Pallets incorrect
102	Protective sleeves missing
103	Strapping loose or missing
104	Marking incorrect
Minor: AQL 10.0 percent defective	
201	Marking illegible

4.2.4 Tests.

4.2.4.1 Filling height. The filled, closed and cleaned projectile shall be radiographed while the WP is molten as specified, and the projectile in a vertical, filled-end-up position.

4.2.4.2 Leakage. After final cleaning, the projectile shall be placed in an oven in a vertical, filling end down position. For the 160° ± 5°F method and the alternate method, at 207°F to 212°F, the projectile shall be maintained at the specified temperature for the required period of time and observed for leakage. When tested at 207°F to 212°F, the munition temperature shall be measured by thermocouples inserted in the burster wells of at least two munitions located in the center of each oven bay batch. The emission of smoke constitutes leakage.

5. PACKAGING

5.1 Packing, level C. The filled projectiles shall be packed for shipment in the same manner as the empty projectiles were received.

5.2 Marking. In addition to any special marking required by the contract or order, the containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and any special marking required.

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6.2 Caution. Due to the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling projectiles.

6.3 Filling aid. Projectiles filled with approximately 0.43 pound of WP have been found to comply with 3.4. This information is furnished as an aid to acquisition and not as a criterion for filling.

6.4 Decontamination. The filled projectiles have been cleaned by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well has been effected with a steam lance and a power driven absorbent swab. This information is furnished as an aid to production and not as a production line design criterion.

6.5 Subject term (key word) listing.

Burster
Projectile
Smoke
White phosphorous

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1310-A429

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-11635C(EA)	PROJECTILE, 57 MM, SMOKE, WP, M308A1; FILLING ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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